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Research article

A novel CFFOPI-FOPID controller for AGC performance enhancement of single and multi-area electric power systems

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ABSTRACT

The modern power system (PS) is an intricate manmade control system. Automatic generation control (AGC) executes an imperative contribution in PS operation to preserve its frequency and tieline power flow within a tolerable range following continuously varying load demands. Hence, a wholesome and expert controller for AGC is obligatory to deliver superior power to the end users. The objective of this article is to design a cascaded fuzzy fractional order (FO) PI-FOPID (CFFOPI–FOPID) controller as a novel control strategy for AGC problem solution in electric PS. The controller parameters are determined employing a recent stochastic imperialist competitive algorithm. The controller assures the convergence of deviation in area frequency and tieline power under load disturbances to zero in minimum definite time. Single and interconnected multi-area thermal PS models are utilized as test systems to authorize the efficacy of the method. Results analysis reveals the advantage of CFFOPI–FOPID controller over various prevalent controllers. The models are also explored with appropriate generation rate constraint. Robustness test validates controller's robustness at large changes in system parameters, random change in power demand and additional imperative nonlinearities.

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1. Introduction

The contemporary power system (PS) is an extremely multifarious and tentative system because of numerous disturbances occurring continuously in PS. With the increasing size, altering structures, random load perturbations and complexity of isolated/interconnected PS, automatic generation control (AGC) has turned into a noteworthy criterion in electric PS design, operation and control. AGC balances the power demand and generated power plus losses in a PS. If this balance is slackened, the frequency and tieline power flows between neighboring control areas of PS get departed from their nominal or scheduled values [1]. The control objective of AGC is to curtail these deviations in a minimum possible time during dynamic conditions. In the absence of an effective AGC action, these departures may initiate a series of acute difficulties and even damage the stability and security of PS. A conventional control theory based fixed gain controller may not attain the sought-after performance in all operating conditions [2]. Hence, an effective, sturdy, reliable and intelligent control strategy is indispensable for AGC. Additionally, the design of AGC controller without considering the chief physical constraints such as generation constraint (GRC), time delay (TD) and deadband nonlinearity in governor (GDB) would offer optimistic outcomes and not real ones [2,3].

Conventional PI and PID configured controllers are still the standard tools in industry due to their reliability and simple realization/implementation [3,4], but, generally they do not work properly in changeable conditions. To solve this problem, better performance might be conjectured from the robust control techniques like optimal control [1], H-infinity control [2,3], internal model control [4] etc., though, most of these methods necessitate either evidence of the states of the system or an effective state reconstructor, therefore, might be problematic to implement in real scenario. The inspection of literature on AGC indicates that many new control techniques have been considered by several researchers in the past decade such as gray wolf optimization based integral (I) [5], hybrid adaptive gbest gravitational search-pattern search technique tuned PID [6], krill herd (KH) algorithm tuned PIDD [7], ICA tuned PID [8], wind driven optimization (WDO) algorithm tuned PID with filter (PIDF) [9], firefly algorithm (FA) tuned I/PI/PID [10] controllers etc., and they perform a crucial role in AGC operation. However, in pragmatic PS implementations, they show certain deficiencies.

To ameliorate the performance of conventional I/PI/PID, fractional order I (FOI), FOPI and FOPID controllers are suggested [11–13]. In FOI/FOPI/FOPID, the fractional calculus theory (FCT) is transported into I/PI/PID controllers to mend their outcomes. FO controllers have additional tuning parameters which provide extra flexibility to them. In the past few years, FO controllers have been applied in PS AGC problem resolution and demonstrated advantages over conventional controllers. Some of their implementations are FA optimized FOI/FOPI/2DOF–FOPID [11], genetic

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Nomenclature

P_r	area rating, MW
P_n	initial loading, MW
α_{ij}	area size ratio ($\alpha_{ij} = -P_{ri}/P_{rj}$)
F^0	nominal system frequency, Hz
R	regulation parameter of speed governor, Hz/puMW
β	frequency response characteristics, puMW/Hz
K_{PS}	PS gain
T_{PS}	PS time constant, s
T_G	time constant of speed governor, s
T_T	thermal turbine time constant, s
K_r	reheater gain
T_r	reheater time constant, s
T_{12}	tieline synchronizing coefficient, puMW/rad
ΔP_g	variation in area generation, puMW

algorithm (GA) [12]/gases Brownian motion optimization [13] optimized FOPID and sine cosine algorithm optimized cascade FOPI–FOPID [14,15] controllers.

It is witnessed that fuzzy logic controller (FLC) advances the recital of optimal conventional controller via managing the changes in the system parameter/operating point by online revising the parameters of the controller [16–22]. Here, ICA [16–18]/bacterial foraging optimization algorithm (BFOA) [19]/particle swarm optimization (PSO) [20]/hybrid differential evolution-PSO (DEPSO) [21]/mine blast algorithm [22] optimized fuzzy PI/PID/PIDF-II/PIDF-(1+PI) structured controllers perform superbly in AGC of various traditional/restructured PSs in comparison to their optimized conventional equivalents. Further, FCT has also been amalgamated with FLC to improve its potential such as FO-fuzzy-PD+I controller optimized by black hole optimization algorithm [23], ICA optimized FOFPID [24]/PIDN-FOI [25]/FFOPI-FOPD [26], BFOA [27,28]/chaotic PSO [29]/quasi-oppositional HS algorithm [30]/biogeography-based algorithm [31] optimized FOFPID controllers. Recently, while researches are being allied with FO and cascaded FO controllers, amalgamation of these notions with FLC is still not been taken care of. Consequently, attention on cascade fuzzy FO controller in AGC studies is needed to be emphasized. Hence, prompted by the achievements attained by intelligence based FO fuzzy controllers and cascade FOPI–FOPID/FFOPI-FOPD controllers designed in [14,15,26], in this paper, a novel cascade fuzzy FOPI–FOPID (CFFOPI–FOPID) controller is offered for single and interconnected 3-area thermal systems.

The main advantages of imperialist competitive algorithm (ICA) compared to other evolutionary optimization technique are its immense convergence rate, robust behavior and tendency not to trap in local minima. Hence, ICA due to its better outcome in PS as demonstrated in [8,16–18,24–26] is employed here to perform the task of controller optimization. The proposed controller structure is simple, easy to realize and assures the robust conduct regardless of the existing uncertainties occurred due to parameter deviations and random demand perturbations in AGC systems. A juxtaposition with the prevailing techniques signifies the dividends of the suggested controller. In comparison with the author' previous works, the present work is a maiden use of cascade fuzzy FOPI–FOPID controller for AGC performance enrichment of single and interconnected power systems.

Considering the above discussions, the foremost motives, novelties and contributions that can be devised for the proposed control scheme are given as:

(a) To offer an optimized CFFOPI–FOPID structured controller with FFOPI and FOPID controllers as master and slave controllers, respectively for AGC of single and 3-area power systems.
(b) To corroborate the preeminence of CFFOPI–FOPID controller over different controllers prevalent in the literature.
(c) To inspect the impact of GRC on the conduct of the systems.
(d) To execute robustness analysis of CFFOPI–FOPID controller for extensive variation in the system parameters, random SLP and additional nonlinearities.

The rest of the paper is categorized as follows: Studied PS models are explicated in Section 2. Then, the suggested control scheme is defined in Section 3. The problem formulation is discussed in Section 4. Then, in Section 5, overview of ICA is provided. The comprehensive simulation results are given and analyzed in Section 6. A robustness analysis is conducted in Section 7. Finally, Section 8 delivers the final observations of the work.

2. Power system models

Investigations are performed on 1-area non-reheat thermal and interconnected unequal area 3-area reheat thermal PSs. 1-area PS comprises of 1 non-reheat thermal generating unit as represented in Fig. 1(a). However, 3-area PS consists of 1 single reheat type thermal unit in its each area (Fig. 4). Each PS area contains governor, turbine, load and generator. The area rating of 1-area PS is 2000 MW. Though, the capacities of 1, 2 and 3 areas of 3-area PS are 2000, 5000 and 8000 MW, respectively. The PS models are prevalent in [1,12]. The per unit (pu) values of parameters of unequal areas are contemplate identical on the respective bases. Thus, while modeling control areas of different capacities, area size ratios for 3-area system are taken as $\alpha_{12} = -P_{r1}/P_{r2}$, $\alpha_{23} = -P_{r2}/P_{r3}$ and $\alpha_{13} = -P_{r1}/P_{r3}$, where all areas are interconnected with each other.

To facilitate the frequency domain study, the transfer functions (TFs) are employed to model components of PS. The TFs of various components used in thermal systems are given below:

The generator TF [4,6,19] is given by

$$G_{PS}(s) = \frac{K_{PS}}{1 + sT_{PS}}. \quad (1)$$

The thermal speed governor TF [4,6,9,19] is given by

$$G_{TG}(s) = \frac{1}{1 + sT_G}. \quad (2)$$

The steam turbine TF [4,6,9,19] is given by

$$G_{TT}(s) = \frac{1}{1 + sT_T} \quad (3)$$

The reheater TF [4,19] is given by

$$G_{TTr}(s) = \frac{1 + sK_rT_r}{1 + sT_r}. \quad (4)$$

The complete mathematical modeling of 1 and 3-area PSs is provided in [1,12]. The parameters of PSs are utilized from [1,10,12] and given in Appendix.

3. Cascade FFOPI–FOPID controller

Regulating the two successive processes having two-loops gives the theory of cascade control (CC). It offers advanced control

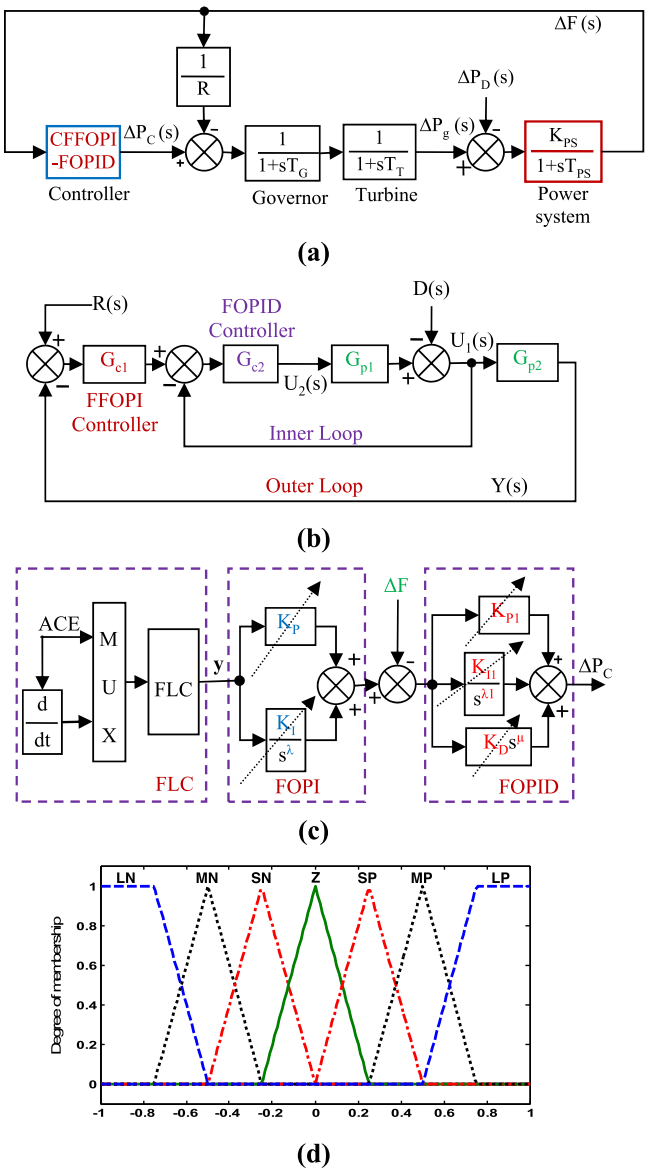


Fig. 1. Different components of the system. **(a)** TF model of 1-area system [12, 26], **(b)** block diagram of the cascade controller, **(c)** CFFOPI–FOPID controller structure and **(d)** MFs for error, error derivative and output of FLC [25,26].

over 1-loop control. A simple CC with two-loops is indicated in Fig. 1(b). The output of the inner loop supplies the outer process/loop. CC includes the usage of two controllers where first controller's output is the set point for the second controller. CC technique is used to minimize load disturbance impact and avoids its spread to other parts of the system. Contrasting to conventional, CC consists of 2 loops termed as slave/secondary/inner loop and master/outer/primary loop. The inner loop replies quite quicker compared to the outer loop. The main share of disturbance is eliminated via the inner loop and the rest is rejected via outer loop. CC also provides improved performance in nonlinear control systems [14,15]. In Fig. 1(b), D (s) and R(s) are the disturbance and reference inputs to the plant, in order. G_{c1} (s)/G_{c2} (s) represent TFs of master/slave controllers, G_{p1} (s)/G_{p2} (s) are TFs of inner/outer processes and U_1 (s)/U_2 (s) are the inputs to the outer/inner processes, in order. Y (s) is the controlled output of the system.

The FO controllers (FOCs) are resourceful enough to govern the dynamical systems efficiently; hence, the notion of cascading the FOCs can additionally improve performance of the system. The concept of FOC deals with differential equations via fractional calculus. The frequently practiced description for fractional integral and derivative is given by the Riemann–Liouville (R–L) definition [14,15]. The system defined via differential equations with fractional derivatives creates TFs with FO of s by means of Laplace transformation. Present study utilizes the CRONE approximation offered by Oustaloup [32]. It employs a recursive distribution of N number of poles and zeros, giving a TF having the pre-designated frequency limit (ω_L, ω_H). In this study, N, ω_L and ω_H are equal to 6, 10^{-2} and 10^2, respectively. The FOPI/FOPID i.e., PI^{λ}/$PI^{\lambda}D^{\mu}$ is the naivest form of FO controller. The TFs of PI, FOPI and FOPID controllers are given by Eqs. (5)–(7).

$$G_{PI}(s) = K_P + \frac{K_I}{s} \text{ (PI)} \tag{5}$$

$$G_{FOPI}(s) = K_P + \frac{K_I}{s^{\lambda}} \text{ (FOPI)} \tag{6}$$

$$G_{FOPID}(s) = K_P + \frac{K_I}{s^{\lambda}} + K_D s^{\mu} \text{ (FOPID)} \tag{7}$$

The conventional PI controller has two parameters (K_P and K_I), FOPI controller has three parameters (K_P, K_I and λ) and FOPID controller has five independent knobs or parameters (K_P, K_I, λ, K_D and μ) to be tuned. Here, FO operators λ and μ are the non-integer order of integrator and differentiator, respectively and may be real numbers. In Eq. (6), if $\lambda = 1$, FOPI will become conventional PI controller. Similarly, in Eq. (7), if $\lambda = \mu = 1$, FOPID will become conventional PID controller. FOPI and FOPID controllers offer more degrees of freedom, superior results having less overshoots/undershoots/oscillations/settling time and great robustness against change in parameters/disturbances/nonlinearities [11–13]. Motivated by these benefits of FOPI/FOPID, in this study, FOPI and FOPID are incorporated in the suggested cascade controller.

Fuzzy controller (FC) is less sensitive to parametric changes and exhibits great vigor than the conventional controllers in controlling system output. The FC uses two inputs i.e., ACE (area control error) and ACE derivative. Membership functions (MFs) of triangular shape are roped in for five inner and trapezoidal for two outer fuzzy linguistic variables such as large negative (LN), medium negative (MN), small negative (SN), zero (Z), small positive (SP), medium positive (MP) and large positive (LP) for inputs/output of FC as indicated in Fig. 1(d). Mamdani FIS is utilized and FC output is framed via center of gravity defuzzification technique. The 2-dimensional rule base for FC inputs/output is utilized from [25,26]. For 1-area and 3-area PSs, ACEs are specified as follows:

$$\text{ACE} = \Delta F \text{ (1-area PS)} \tag{8}$$

$$\left.\begin{aligned} \text{ACE}_1 &= \beta_1 \Delta F_1 + \Delta P\text{tie}_1 \\ \text{ACE}_2 &= \beta_2 \Delta F_2 + \Delta P\text{tie}_2 \\ \text{ACE}_3 &= \beta_3 \Delta F_3 + \Delta P\text{tie}_3 \end{aligned}\right\} \text{ (3-area PS)} \tag{9}$$

where, $\beta_1/\beta_2/\beta_3$ are the frequency response characteristics, $\Delta F/\Delta F_1/\Delta F_2/\Delta F_3$ are the deviation in the area frequencies, $\Delta P\text{tie}_1/\Delta P\text{tie}_2/\Delta P\text{tie}_3$ are the deviation in the tieline powers and $\Delta P\text{tie}_1$ is the sum of $\Delta P\text{tie}_{12}$ and $\Delta P\text{tie}_{13}$. $\Delta P\text{tie}_2 = \alpha_{12}\Delta P\text{tie}_{12} + \Delta P\text{tie}_{23}$ and $\Delta P\text{tie}_3 = \alpha_{13}\Delta P\text{tie}_{13} + \alpha_{23}\Delta P\text{tie}_{23}$, where, $\alpha_{12}/\alpha_{13}/\alpha_{23}$ are the area rating ratios.

Induced with the above discussions, in this study, the notion of CC alongside with fractional calculus and fuzzy is amalgamated to propose a novel cascade fuzzy FOPI–FOPID (CFFOPI–FOPID) controller. The controller incorporates the benefits of CC, fuzzy logic and FO controllers. In cascade controller of Fig. 1(b), the G_{c1} and G_{c2} controllers are given by Eqs. (10) and (11), respectively.

$$G_{c1}(s) = yK_P + \frac{yK_I}{s^{\lambda}} \text{(FFOPI)} \tag{10}$$

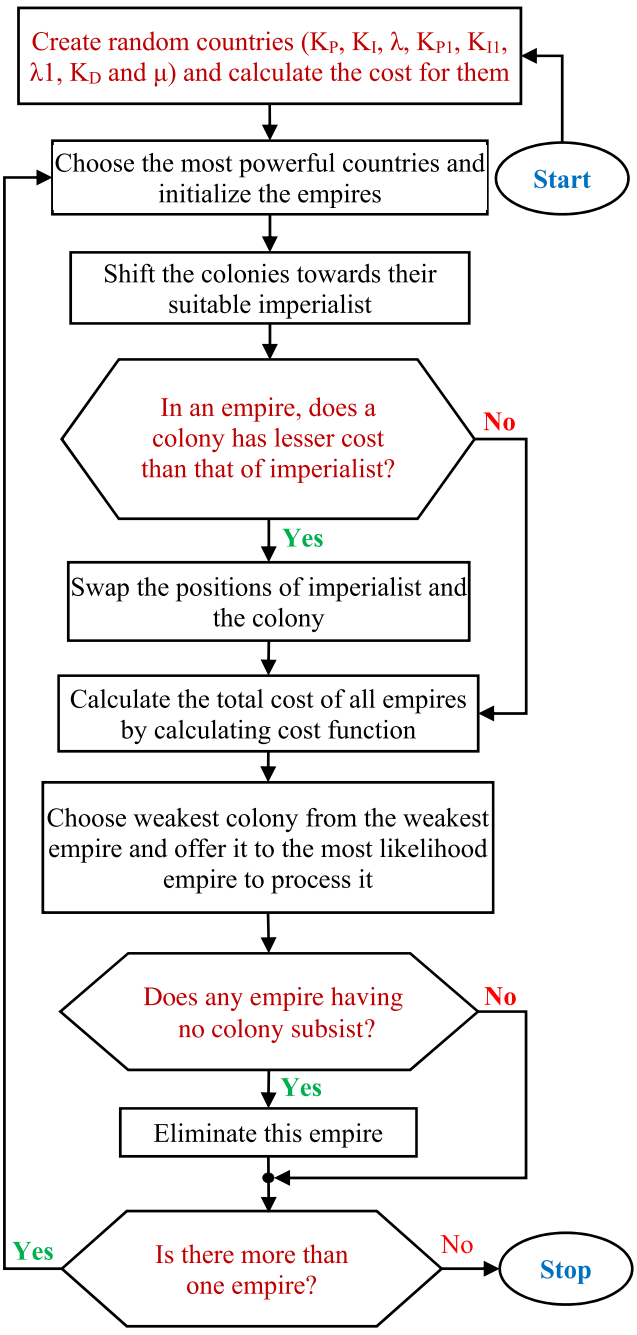


Fig. 2. ICA flow chart for CFFOPI–FOPID controller.

$$G_{c2}(s) = K_{P1} + \frac{K_{I1}}{s^{\lambda 1}} + K_D s^{\mu}(\text{FOPID}) \tag{11}$$

where, y is the output of FC acting as the input to FOPI controller. The organization of the CFFOPI–FOPID controller is displayed in Fig. 1(c). It has eight parameters (K_P, K_I, λ, K_{P1}, K_{I1}, $\lambda 1$, K_D and μ) to be optimized. Identical CFFOPI–FOPID controllers are employed in each area of interconnected PS. The output of CFFOPI–FOPID controller is the control input i.e. ΔP_C for the PSs as represented by Eq. (12).

$$\Delta P_C = \left[\left\{\left(yK_P + \frac{yK_I}{s^{\lambda}}\right) - \Delta F\right\} \times \left(K_{P1} + \frac{K_{I1}}{s^{\lambda 1}} + K_D s^{\mu}\right)\right] \tag{12}$$

4. Problem formulation

To accomplish generation-load balance via AGC, the CFFOPI–FOPID controller must be optimized so as to underplay an objective function (J). The choice of the J plays a crucial role to boost the system dynamic results. Also, the demonstration of an optimization technique rests on the selection of an appropriate J. In industry for the design of controllers, the frequently used objective functions (J_S) in time domain are IAE (integral of absolute error), ITAE (time multiplied IAE), ISE (integral of squared error) and ITSE (time multiplied ISE), [16,17,19,25–28,30]. Due to superior performance, in the present work, ISE defined via (13) is exploited to tune the proposed controller. The other J_S for 3-area system are defined using Eqs. (14)–(16).

$$\text{ISE} = \int_0^{\infty} \left\{\Delta F_1^2 + \Delta F_2^2 + \Delta F_3^2 + \Delta \text{Ptie}_1^2 + \Delta \text{Ptie}_2^2 + \Delta \text{Ptie}_3^2\right\} dt \tag{13}$$

$$\text{ITSE} = \int_0^{\infty} t\left\{\Delta F_1^2 + \Delta F_2^2 + \Delta F_3^2 + \Delta \text{Ptie}_1^2 + \Delta \text{Ptie}_2^2 + \Delta \text{Ptie}_3^2\right\} dt \tag{14}$$

$$\text{IAE} = \int_0^{\infty} \{|\Delta F_1| + |\Delta F_2| + |\Delta F_3| + |\Delta \text{Ptie}_1| + |\Delta \text{Ptie}_2| + |\Delta \text{Ptie}_3|\} dt \tag{15}$$

$$\text{ITAE} = \int_0^{\infty} t\{|\Delta F_1| + |\Delta F_2| + |\Delta F_3| + |\Delta \text{Ptie}_1| + |\Delta \text{Ptie}_2| + |\Delta \text{Ptie}_3|\} dt \tag{16}$$

Consequently, J_S for 1-area system can be developed. To enhance PS dynamic performance, it is obligatory to curtail J_S during tuning the controller gains. The CFFOPI–FOPID controller optimization task is specified as the subsequent constrained optimization problem. Minimize J subject to:

$$\left.\begin{array}{l} K_P^{\min} \le K_P \le K_P^{\max} \\ K_I^{\min} \le K_I \le K_I^{\max} \\ \lambda^{\min} \le \lambda \le \lambda^{\max} \\ K_{P1}^{\min} \le K_{P1} \le K_{P1}^{\max} \\ K_{I1}^{\min} \le K_{I1} \le K_{I1}^{\max} \\ \lambda 1^{\min} \le \lambda 1 \le \lambda 1^{\max} \\ K_D^{\min} \le K_D \le K_D^{\max} \\ \mu^{\min} \le \mu \le \mu^{\max} \end{array}\right\} \tag{17}$$

In (17), min and max represent the minimum and maximum bounds of the controller parameters.

5. Imperialist competitive algorithm (ICA)

The evolutionary techniques recognize well established accomplishments over the customary gradient based techniques encountered in control of electric PSs. ICA is a modern metaheuristics evolutionary optimization method which represents the social and political developments of imperialism and imperialistic competition [33]. It imitates the political/social demeanor of imperialist countries in an endeavor to govern the feeble countries. ICA commences with a random population of solutions termed as countries. As per their powers, the countries divide into various imperialists and colonies which together creates some empires. Every empire comprises of one imperialist and some colonies. The imperialistic competition (IC) amongst empires establishes the origin of ICA. These empires begin to grow concurrently by moving all colonies toward the imperialists through an operation called assimilation. During IC, feeble empires breakdown and influential ones take ownership of their colonies. IC finally meets to a condition in where only one empire

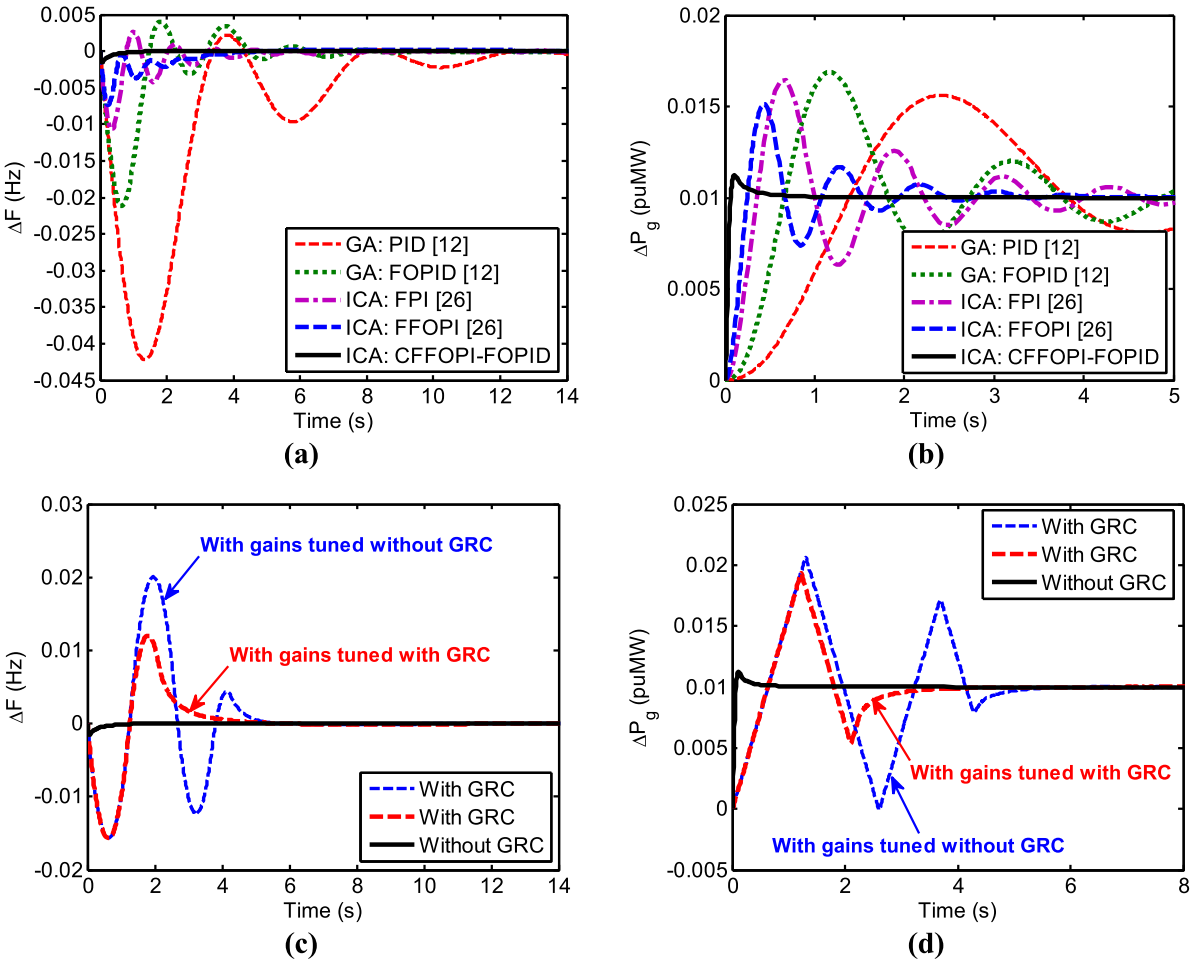


Fig. 3. 1-area system results at $\Delta P_D = 0.01$ puMW. **(a)** ΔF, **(b)** ΔP_g, **(c)** ΔF with CFFOPI–FOPID with/without GRC and **(d)** ΔP_g with CFFOPI–FOPID with/without GRC.

with its colonies have the identical position and cost as the imperialist [33].

Compared to other metaheuristics algorithms, ICA has various benefits such better convergence rate, well-suited for dissimilar optimization problems, strong neighborhood search ability, less dependency on initial solutions, minor likelihoods of trapping into local minimas, authoritative in governing more parameters etc. [34]. Additionally, ICA converges at a faster rate, requires less computational time and offers less computational cost. Hence, computational burden of ICA is less compared to other algorithms [34,35]. Encouraged by these benefits, ICA is utilized in this study to tune parameters of the suggested controller. In this study, only the overview of ICA is presented. For more details, readers may refer [8,16–18,24–26,33–35]. However, flow chart of ICA has been provided in Fig. 2. The parameters of ICA are chosen as [16]: iteration numbers = 100, countries numbers = 500, initial empires = 40; initial colonies = 460; revolution rate = 0.3, constant related to the average power of empire's colonies = 0.05, assimilation constant = 1.5 and assimilation angle constant = 0.5.

6. Simulation results

6.1. 1-area PS

Analogous to [12], the system dynamics of 1-area PS with non-reheat thermal unit revealed in Fig. 1(a) is evaluated in SIMULINK/MATLAB atmosphere against 1% SLP (step load perturbation) implemented at time = 0 s. The applicable data utilized in simulations are provided in Appendix. To handle the perturbation, CFFOPI–FOPID controller is designed employing ICA by minimizing J_{ISE}. The optimized gains of CFFOPI–FOPID controller along with published FPI and FFOPI controllers are tabulated in Table 1. To illustrate the controller efficiency, the closed-loop results attained due to CFFOPI–FOPID controller are contrasted to the existing results of ICA optimized FPI/FFOPI [26] and genetic algorithm (GA) optimized PID/FOPID [12] controllers. The system time-domain responses for variation in area frequency (ΔF) and variation in generated power (ΔP_g) are compared in Fig. 3(a–b). The transient characteristics like peak overshoot (PO), peak undershoot (PO), settling time (ST) and J_S (ISE, ITSE, IAE and ITAE) of the system responses are computed and given in Table 2. From Table 2 and Fig. 3(a–b), it is remarkable that ICA tuned CFFOPI–FOPID controller is excellent compared to ICA optimized FFOPI/FPI [26] and GA optimized PID/FOPID [12] controllers regarding less ST, PO, PU, number of oscillations and J_S. It is exposed from Table 2 that, lowest J_S (ISE = 4.68×10^{-7}; ITSE = 1.34×10^{-7}; IAE = 8.51×10^{-4}; ITAE = 1.86×10^{-3}), ST ($\Delta F = 0.40$ s; $\Delta P_g = 0.25$ s), US ($\Delta F = -0.0015$ Hz; $\Delta P_g = 0$ puMW) and OS ($\Delta F = 0$ Hz; $\Delta P_g = 0.0112$ puMW) values are obtained with CFFOPI–FOPID controller compared to FPI/FFOPI [26] and PID/FOPID [12] structured controllers. Hence, ICA: CFFOPI–FOPID controller delivers better performance in comparison to other techniques. However, the performance of CFFOPI–FOPID controller gets deteriorated when an important constraint called

Table 1
ICA optimized FPI/FFOPI/CFFOPI–FOPID controller parameters.

Parameters	1-area system			1-area system (GRC)	3-area system			3-area system (GRC)
	FPI [26]	FFOPI [26]	CFFOPI–FOPID	CFFOPI–FOPID	FPI [26]	FFOPI [26]	CFFOPI–FOPID	CFFOPI–FOPID
K_P	0.8621	1.9058	0.6579	0.9896	0.9478	1.4412	1.5324	2.9526
K_I	1.3969	1.2685	2.6341	2.5689	0.9145	1.5955	1.8526	1.7568
λ	–	1.1358	0.9481	0.9281	–	0.9879	0.5351	0.4564
K_{P1}	–	–	1.6827	1.5368	–	–	1.3527	2.6524
K_{I1}	–	–	2.8864	2.7930	–	–	1.8564	1.5627
$\lambda 1$	–	–	0.4867	0.4657	–	–	0.5321	0.4851
K_D	–	–	1.5537	1.9324	–	–	0.2631	1.8864
μ	–	–	0.9572	0.8624	–	–	0.7529	0.5614

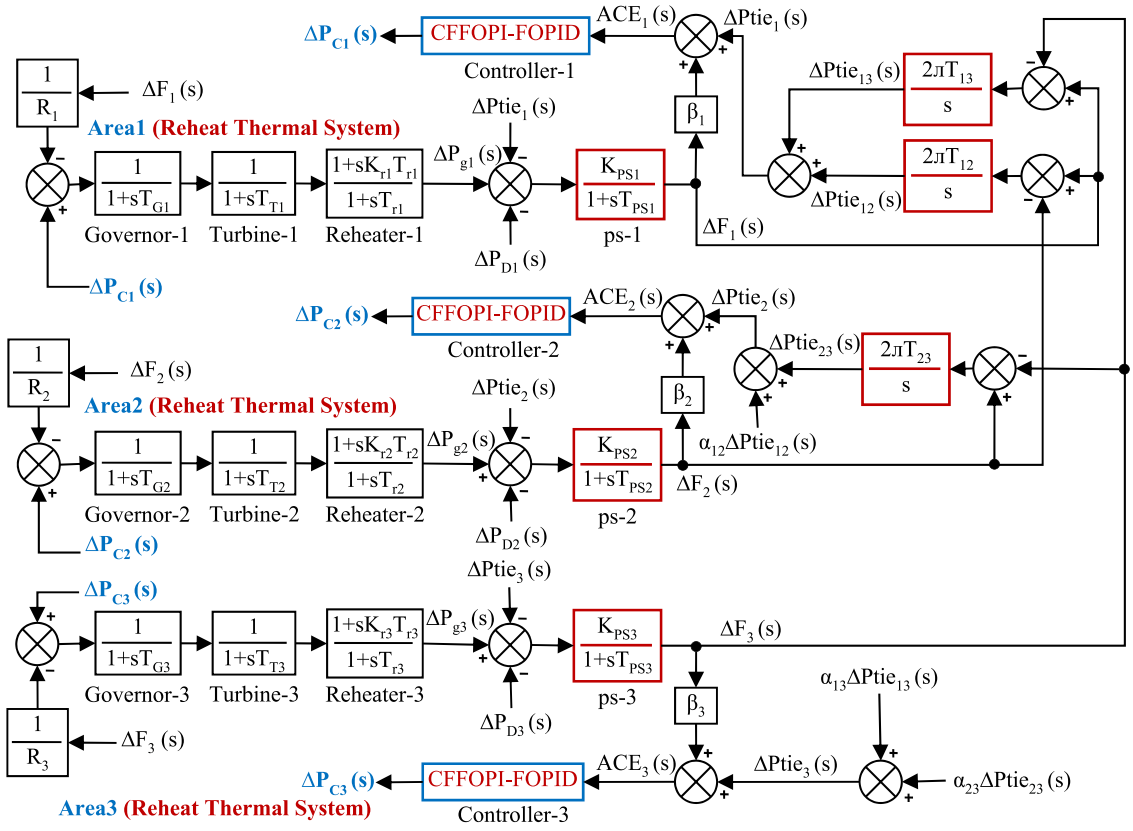


Fig. 4. Interconnected 3-area PS TF model [1,26].

GRC (generation rate constraint) is considered. Because of the limit of thermal/mechanical movements in thermal plants, generation can fluctuate simply at an indicated highest rate termed as GRC. The realistic AGC studies should incorporate the GRC effect. The closed loop GRC limit of ±0.008 pu/s is taken here. GRC restricts the instant response of the PS to reject disturbance. Hence, system results get corrupted as indicated in Fig. 3(c–d) and Table 2 where the values of J_S (ISE = 5.64×10^{-4}; ITSE = 1.02×10^{-3}; IAE = 4.43×10^{-2}; ITAE = 9.10×10^{-2}), ST (ΔF = 5.02 s; ΔP_g = 4.77 s), US ($\Delta F = -0.0157$ Hz; $\Delta P_g = -6.15 \times 10^{-5}$ puMW) and OS ($\Delta F = 2.01 \times 10^{-2}$ Hz; ΔP_g = 0.0206 puMW) are increased in the existence of GRC and CFFOPI–FOPID controller. However, when CFFOPI–FOPID controller is optimized in the presence of GRC, these values get decrease to J_S (ISE = 2.59×10^{-4}; ITSE = 2.83×10^{-4}; IAE = 2.62×10^{-2}; ITAE = 4.12×10^{-2}), ST (ΔF = 4.09 s; ΔP_g = 2.92 s), US ($\Delta F = -0.0157$ Hz; $\Delta P_g = 5.36 \times 10^{-3}$ puMW) and OS ($\Delta F = 1.20 \times 10^{-2}$ Hz; ΔP_g = 0.0193 puMW). In this scenario, PS regains the balance of generation and demand plus losses in less time having lower overshoot/undershoot. Therefore, GRC must be utilize with optimized controller in AGC schemes to eradicate injurious impacts of GRC.

6.2. 3-area PS

To validate the achievement of the proposed ICA optimized CFFOPI–FOPID controller in interconnected multi-area PS, the work is stretched to an unequal interconnected 3-area PS having reheat thermal units in each area as displayed in Fig. 4 [1]. The power ratings of area-1/2/3 are 2000/5000/8000 MW, respectively. A demand of SLP = 2% is considered in area-1 and ICA optimized parameters of CFFOPI–FOPID and existing FFOPI/FPI controllers are offered in Table 1. Some system results like $\Delta F_1/\Delta F_2/\Delta Ptie_1/\Delta Ptie_3$ due to the proposed CFFOPI–FOPID and existing ICA optimized FFOPI/FPI [26], optimal [1]/KHA [7] optimized PI, FA optimized PID [10], WDO optimized PIDF [9] and DEPSO optimized FPID [21] controllers are shown in Fig. 5(a–d) and in Table 3. It is revealed from the results displayed in Fig. 5(a–d) that ICA: CFFOPI–FOPID controller compared to others offers improved damping effect against system oscillations and swiftly reinstates the system stability by converging the responses to zero steady state value. It is noteworthy from Table 3 that CFFOPI–FOPID controller delivers least J_S, OS, US and fast ST compared to optimal: PI [1], KHA: PI [7], FA:

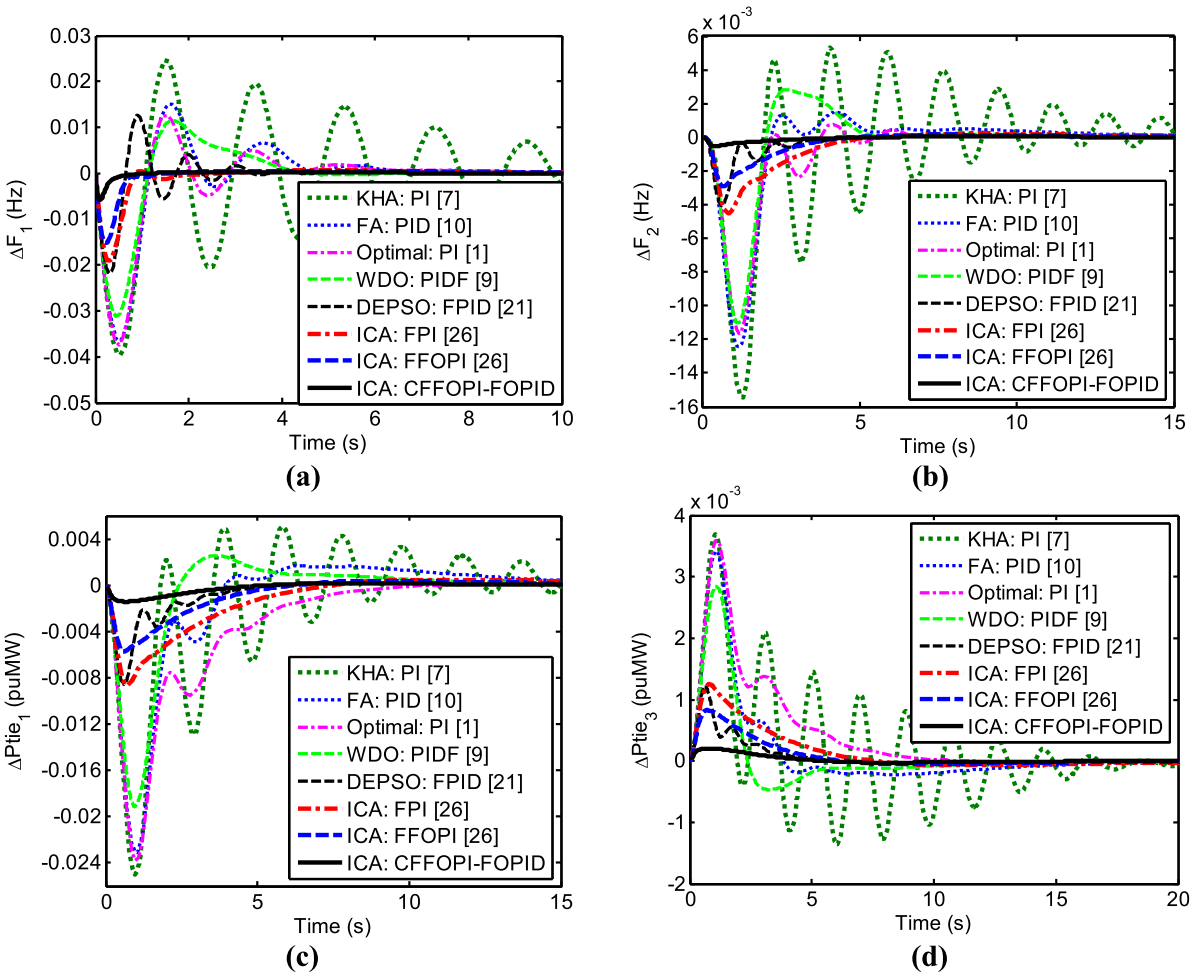


Fig. 5. 3-area system results at $\Delta P_D = 0.02$ puMW. **(a)** ΔF_1, **(b)** ΔF_2, **(c)** $\Delta Ptie_1$ and **(d)** $\Delta Ptie_3$.

Table 2
ST/US/OS/J_S results of 1-area nonreheat thermal PS.

Controller type	ΔF			ΔP_g			J_S			
	ST (s)	US (−ve) (Hz)	OS (Hz)	ST (s)	US (puMW)	OS (puMW)	ISE	ITSE	IAE	ITAE
GA: PID [12]	14.88	0.0422	2.22×10^{-3}	7.83	8.08×10^{-3}	0.0156	2.80×10^{-3}	4.72×10^{-3}	1.08×10^{-1}	3.03×10^{-1}
GA: FOPID [12]	7.16	0.0212	4.05×10^{-3}	5.43	7.47×10^{-3}	0.0169	3.27×10^{-4}	2.67×10^{-4}	2.76×10^{-2}	4.77×10^{-2}
ICA: FPI [26]	4.25	0.0107	2.65×10^{-3}	4.39	6.31×10^{-3}	0.0164	5.39×10^{-5}	3.58×10^{-5}	1.09×10^{-2}	2.14×10^{-2}
ICA: FFOPI [26]	3.26	0.0074	2.40×10^{-4}	2.30	7.39×10^{-3}	0.0151	2.45×10^{-5}	1.93×10^{-5}	8.67×10^{-3}	2.13×10^{-2}
ICA: CFFOPI–FOPID	**0.40**	**0.0015**	**0**	**0.25**	**0**	**0.0112**	**4.68×10^{-7}**	**1.34×10^{-7}**	**8.51×10^{-4}**	**1.86×10^{-3}**
ICA: CFFOPI–FOPID (GRC)	5.02	0.0157	2.01×10^{-2}	4.77	-6.15×10^{-5}	0.0206	5.64×10^{-4}	1.02×10^{-3}	4.43×10^{-2}	9.10×10^{-2}
ICA: CFFOPI–FOPID (With gains tuned with GRC)	**4.09**	0.0157	**1.20×10^{-2}**	**2.92**	**5.36×10^{-3}**	**0.0193**	**2.59×10^{-4}**	**2.83×10^{-4}**	**2.62×10^{-2}**	**4.12×10^{-2}**

Bold designates best values.

PID [10], WDO: PIDF [9], DEPSO: FPID [21], ICA: FPI [26] and ICA: FFOPI [26] controllers. Therefore, ICA: CFFOPI–FOPID acts competently in multi-area electric PS. Fig. 6(a–d) showcases the results of CFFOPI–FOPID for $\Delta F_1/\Delta F_3/\Delta Ptie_1/\Delta Ptie_2$ in the presence of GRC limit of ±0.00167 pu/s. It is observed from the figure that system performance with GRC degrades drastically. Additionally, it is validated from Table 3 in terms of more values of ST/OS/US/J_S for $\Delta F_1/\Delta F_2/\Delta F_3$. Conversely, system results recover on optimization of controller considering the effect of GRC. Hence, the proposed controller establishes sanguine dynamic recital in comparison to other control methods, acts efficiently and guarantees the adequate operation of the system in the company of GRC.

7. Robustness analysis

The robustness of ICA optimized CFFOPI–FOPID controller parameters acquired at the nominal condition of the system is inspected for altered system conditions. In the varied operating conditions, the designed controller should offer desired dynamic performance.

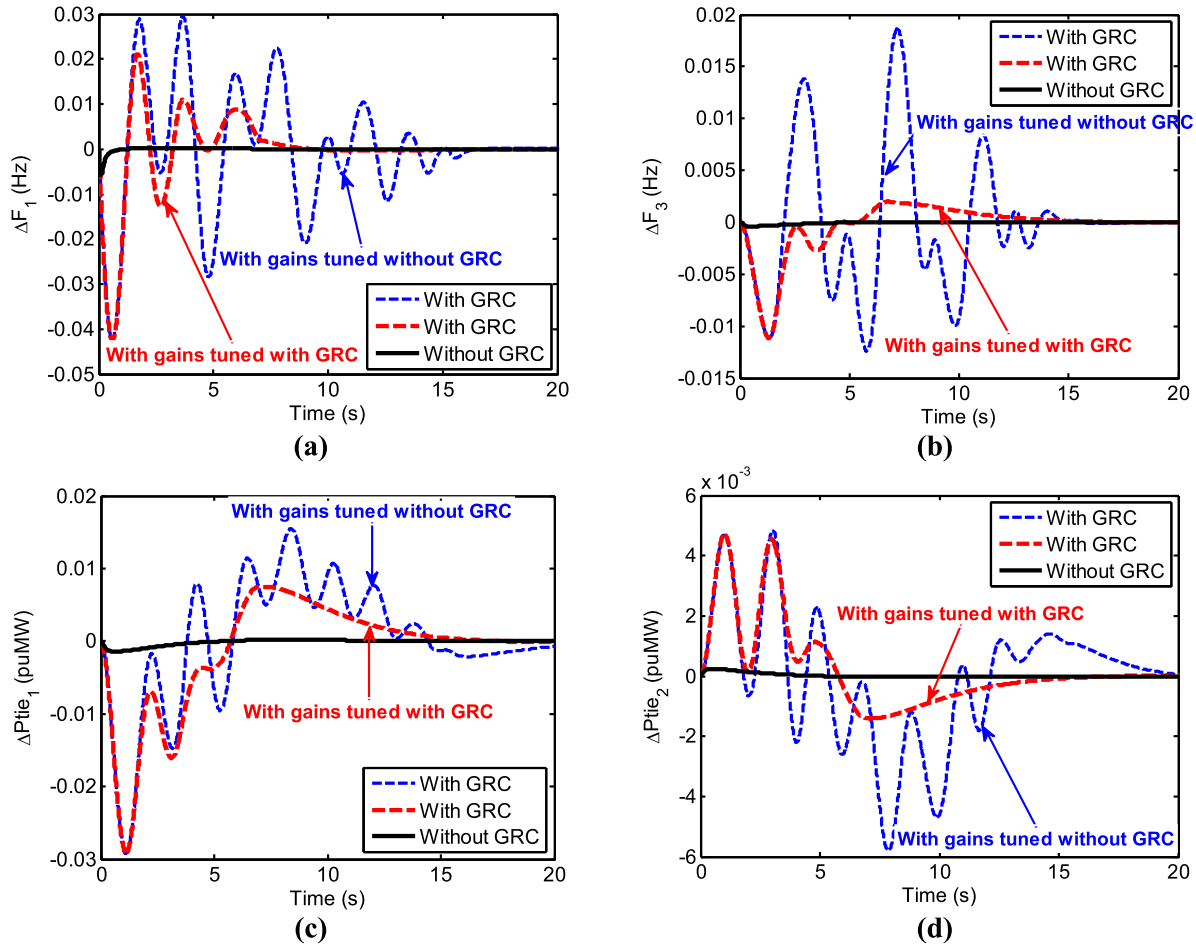


Fig. 6. 3-area system results at $\Delta P_{D1} = 0.02$ puMW with CFFOPI–FOPID with/without GRC. **(a)** ΔF_1, **(b)** ΔF_3, **(c)** $\Delta Ptie_1$ and **(d)** $\Delta Ptie_2$.

Table 3
ST/US/OS/J_S results of 3-area reheat thermal PS.

Controller type	ST (s)			US (−ve) (Hz)		US (−ve) (puMW)	OS (Hz)		OS (puMW)	J_S			
	ΔF_1	ΔF_2	ΔF_3	ΔF_1	ΔF_2	ΔF_3	ΔF_1	ΔF_2	ΔF_3	ISE	ITSE	IAE	ITAE
KHA: PI [7]	24.83	22.61	16.27	0.0397	0.0179	0.0169	0.02460	0.0053	0.0056	2.74×10^{-3}	6.72×10^{-3}	0.2695	1.2100
FA: PID [10]	7.91	5.09	5.34	0.0365	0.0125	0.0111	0.0151	0.0015	0.0014	1.39×10^{-3}	1.39×10^{-3}	0.1311	0.3467
Optimal: PI [1]	7.68	4.39	4.09	0.0374	0.0170	0.0091	0.01270	0.0007	0	1.36×10^{-3}	1.33×10^{-3}	0.1273	0.2596
WDO: PIDF [9]	4.25	4.85	5.02	0.0311	0.0110	0.0088	0.01140	0.0028	0.0029	1.01×10^{-3}	1.00×10^{-3}	0.1079	0.2340
ICA: FPI [26]	6.35	3.87	4.01	0.0191	0.0045	0.0038	0.00059	0	0	2.3×10^{-4}	2.00×10^{-4}	0.0583	0.1718
DEPSO: FPID [21]	4.20	2.92	2.98	0.0218	0.0040	0.0031	0.0126	0	0	2.50×10^{-4}	1.51×10^{-4}	0.0406	0.0701
ICA: FFOPI [26]	4.01	3.10	3.26	0.0153	0.0029	0.0023	0.00051	0	0	1.10×10^{-4}	8.00×10^{-5}	0.0369	0.1017
ICA: CFFOPI–FOPID	**0.62**	**0.54**	**0.66**	**0.0058**	**0.0005**	**0.0004**	**0.00018**	**0**	**0**	$\mathbf{8.00 \times 10^{-6}}$	$\mathbf{5.00 \times 10^{-6}}$	**0.0096**	**0.0280**
ICA: CFFOPI–FOPID (GRC)	16.09	16.90	14.44	0.0424	0.0161	0.0124	0.02950	0.0179	0.0186	6.15×10^{-3}	2.77×10^{-2}	0.4931	2.8650
ICA: CFFOPI–FOPID (With gains tuned with GRC)	**8.41**	**12.02**	**12.15**	**0.0425**	**0.0147**	**0.0112**	**0.02110**	**0.0063**	**0.0019**	$\mathbf{2.57 \times 10^{-3}}$	$\mathbf{4.50 \times 10^{-3}}$	**0.2416**	**0.9075**

Bold designates best values.

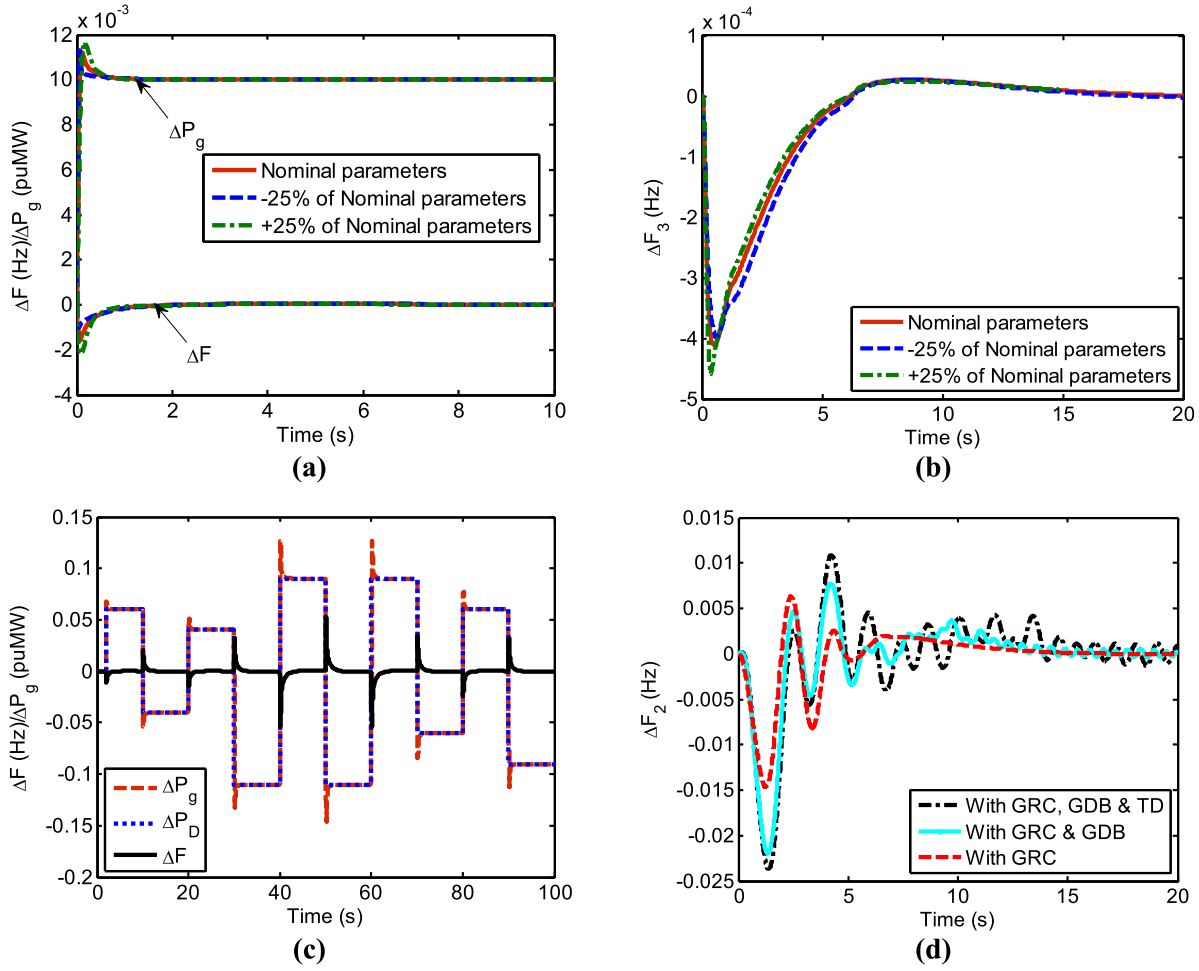


Fig. 7. Robustness analysis results. **(a)** $\Delta F/\Delta P_g$ of 1-area system at $\Delta P_D = 0.01$ puMW, **(b)** ΔF_3 of 3-area system at $\Delta P_{D1} = 0.02$ puMW, **(c)** $\Delta F/\Delta P_g$ of 1-area system at random ΔP_D and **(d)** ΔF_2 of 3-area system with GRC/GDB/TD.

7.1. Variation in system parameters

The controller considered in nominal condition must be robust enough to take care of the variation in system parameters occurred usually in the interconnected PS. The system parameters are varied ±25% from the nominal values. The system dynamic results at nominal and changed conditions are displayed and compared in Fig. 7(a–b) for $\Delta F/\Delta P_g$ (1-area system) and ΔF_3 (3-area system) responses. In Fig. 7(a–b), the responses at nominal and changed parameters are nearly same. Hence, the proposed controller reveals significant robustness to the system parameter variations.

7.2. Random SLP

The designed controller should be sturdy at random load variations taking place in a real PS. Fig. 7(c) displays the comparison of frequency/generation variation versus time for 1-area system under random SLP with the gains acquired at 1% SLP or ΔP_D. It is demonstrable from the results that the offered controller performs robustly and delivers stable performance during RLP.

7.3. Effect of GDB and TD

Unless all major physical constraints like GRC, GDB and TD are incorporated, the designed controller cannot be feasible for the actual environment and the simulated results may be far from the reality. GRC in steam turbine, DB in speed governor and TD in communication channels are the integral parts of a real PS. However, their consideration in simulations produces undesirable results having higher OS/US, longer ST and more number of oscillations. If a controller is not vigorous enough, the system undoubtedly will lose its stability under pragmatic condition of physical constraints. Fig. 7(d) demonstrates ΔF_2 response due to the proposed controller optimized for GRC condition for 3-area system with physical constraints. Here, GRC has ±0.00167 pu/s values [24]. In GDB, $N_1 = 0.8$ and $N_2 = -0.2$ [17]. For TD, 0.015 s transport delay at the input & output of controller and output of 1/R block are employed. Fig. 7(d) plainly divulges that the proposed controller conducts robustly in the company of GRC, both GRC & GDB and combined GRC, GDB & TD nonlinearities. Thus, the proposed ICA optimized CFFOPI–FOPID controller continues the coveted control operation.

8. Conclusion

In this work, a CFFOPI–FOPID controller tuned via ICA is being suggested for power-frequency stabilization of single and multi-area thermal PSs. The significant verdicts of the article can be succinctly summed up as follows:

(1) The proposed ICA: CFFOPI–FOPID controller substantially advances the frequency, power generation and tieline power flow profiles of the single and interconnected 3-area thermal PSs in comparison to ICA: FPI/FFOPI, GA: PID/FOPID, Optimal/KHA: PI, FA: PID, WDO: PIDF and

DEPSO: FPID controllers in terms of fast responses, less oscillations, low undershoots/overshoots, small settling times and minimum values of ISE/ITSE/IAE/ITAE objective functions.

(2) GRC consideration in the thermal turbines impacts the system performance adversely. However, an amelioration in the damping capability of AGC can be achieved by optimizing the controller parameters in the incidence of GRC.

(3) The structure of the designed controller is validated robust under ±25% variation in parameters of the PSs from their nominal values, RLP and physical constraints.

(4) In addition to GRC, the physical constraints like GDB and TD also deteriorate the system outcome. However, controller designed without considering these physical constraints will not be feasible for practical scenario.

The author has intentions to implement the CFFOPI–FOPID control method in future to control the power-frequency and voltage of a real multi-area interconnected PS.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Nominal system parameters

1-Area non-reheat thermal PS [12,26]**:**

$P_r = 2000$ MW, $P_n = 50\%$, $F^0 = 60$ Hz, $R = 3$ Hz/puMW, $T_G = 0.4$ s, $T_T = 0.5$ s, $K_{PS} = 100$, $T_{PS} = 20$ s, $\Delta P_D = 0.01$ puMW.

3-Area reheat thermal PS [1,10,26]**:**

$P_r = 2000$ MW, $P_n = 50\%$, $F^0 = 60$ Hz, $\beta = 0.425$ puMW/Hz, $R = 2.4$ Hz/puMW, $T_G = 0.08$ s, $T_T = 0.3$ s, $K_r = 0.5$, $T_r = 10$ s, $K_{PS} = 120$, $T_{PS} = 20$ s, $T_{12} = 0.0866$ puMW/rad, $\alpha_{12} = -2/5$, $\alpha_{13} = -2/8$, $\alpha_{23} = -5/8$, $\Delta P_{D1} = 0.02$ puMW.

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